**1.** Who proposed that energy is emitted or absorbed in discrete packets called quanta (E = h*nu)?
- **(A)** Albert Einstein
- **(B)** Max Planck (1900)
- **(C)** Niels Bohr
- **(D)** Louis de Broglie

**2.** The failure of classical Rayleigh-Jeans law to explain blackbody spectra at short wavelengths is termed:
- **(A)** Infrared divergence
- **(B)** Photoelectric anomaly
- **(C)** Compton scattering
- **(D)** Ultraviolet catastrophe

**3.** In the photoelectric effect, the maximum kinetic energy of emitted photoelectrons depends strictly on:
- **(A)** Frequency of the incident light
- **(B)** Intensity of incident light
- **(C)** Surface area of target
- **(D)** Duration of illumination

**4.** The minimum energy required to liberate an electron from a metal surface is known as its:
- **(A)** Ionization energy
- **(B)** Fermi energy
- **(C)** Work function (Phi)
- **(D)** Excitation potential

**5.** Compton effect confirmed that X-ray photons possess both energy and:
- **(A)** Linear momentum (p = h / lambda)
- **(B)** Electric charge
- **(C)** Rest mass
- **(D)** Magnetic dipole moment

**6.** Who hypothesized in 1924 that moving material particles exhibit wave-like properties?
- **(A)** Erwin Schrodinger
- **(B)** Werner Heisenberg
- **(C)** Louis de Broglie
- **(D)** Max Born

**7.** The de Broglie matter wavelength lambda associated with a particle of momentum p is:
- **(A)** lambda = p / h
- **(B)** lambda = h / p
- **(C)** lambda = h * p
- **(D)** lambda = h / (2 * p)

**8.** The de Broglie wavelength of an electron accelerated from rest across potential difference V is:
- **(A)** 12.27 / V nm
- **(B)** 0.1227 * sqrt(V) nm
- **(C)** 1.227 * V nm
- **(D)** 1.227 / sqrt(V) nm

**9.** Which historic experiment first experimentally verified the wave nature of electrons in 1927?
- **(A)** Rutherford alpha scattering
- **(B)** Stern-Gerlach experiment
- **(C)** Davisson-Germer experiment
- **(D)** Franck-Hertz experiment

**10.** In the Davisson-Germer experiment, electron waves were diffracted by the atomic planes of a:
- **(A)** Nickel single crystal
- **(B)** Gold foil
- **(C)** Graphite sheet
- **(D)** Silicon wafer

**11.** The phase velocity v_p of a wave with angular frequency omega and wavevector k is given by:
- **(A)** d(omega) / d(k)
- **(B)** k / omega
- **(C)** omega * k
- **(D)** omega / k

**12.** The group velocity v_g of a wave packet is given by:
- **(A)** omega / k
- **(B)** d(omega) / d(k)
- **(C)** d(k) / d(omega)
- **(D)** k * omega

**13.** For a de Broglie wave packet representing a free particle, the group velocity v_g is:
- **(A)** Equal to the speed of light c
- **(B)** Exactly equal to the classical particle velocity v
- **(C)** Half the particle velocity
- **(D)** Twice the phase velocity

**14.** Heisenberg's Uncertainty Principle states that the product of uncertainties in position and momentum satisfies:
- **(A)** Delta_x * Delta_p >= hbar / 2
- **(B)** Delta_x * Delta_p = 0
- **(C)** Delta_x * Delta_p <= hbar
- **(D)** Delta_x * Delta_p >= h

**15.** The reduced Planck constant hbar is defined as:

**(A)** 2 * pi * h
**(B)** h / pi
**(C)** h / (2 * pi)
**(D)** h / 4

**16.** The uncertainty relation connecting energy and time is:

**(A)** Delta_E * Delta_t = 0
**(B)** Delta_E / Delta_t >= hbar
**(C)** Delta_E * Delta_t <= h
**(D)** Delta_E * Delta_t >= hbar / 2

**17.** Why cannot an electron reside permanently inside an atomic nucleus (radius ~ 10^-14 m)?

**(A)** Nuclear forces repel negative charges
**(B)** Electrons have no mass
**(C)** It would annihilate with protons instantly
**(D)** Its kinetic energy by uncertainty principle would vastly exceed nuclear binding energy

**18.** The wave function Psi(x,t) in quantum mechanics represents:

**(A)** A physical electric field
**(B)** A mechanical displacement wave
**(C)** A complex probability amplitude
**(D)** A measurable particle trajectory

**19.** According to Max Born's statistical postulate, the probability density P(x,t) is given by:

**(A)** |Psi(x,t)|^2 = Psi* * Psi
**(B)** Psi(x,t)
**(C)** dPsi / dx
**(D)** Real(Psi)

**20.** The normalization condition for a 1D wave function over all space is:

**(A)** Integral |Psi|^2 dx = 0
**(B)** Integral_{-inf}^inf |Psi(x,t)|^2 dx = 1
**(C)** Integral Psi dx = 1
**(D)** Integral (dPsi/dx) dx = 0

**21.** Which of the following is a mandatory mathematical condition for a well-behaved wave function?

**(A)** It must be purely real
**(B)** It must be continuous, single-valued, and square-integrable everywhere
**(C)** It must diverge to infinity at boundaries
**(D)** It must be non-zero everywhere

**22.** In quantum mechanics, all physically observable quantities are represented by:

**(A)** Scalar numbers only
**(B)** Non-linear tensors
**(C)** Unitary matrices with zero trace
**(D)** Linear Hermitian operators

**23.** The linear momentum operator p_hat in the 1D position representation is:

**(A)** - i * hbar * (d / dx)
**(B)** + i * hbar * (d / dx)
**(C)** - hbar^2 * (d^2 / dx^2)
**(D)** hbar * (d / dt)

**24.** The total energy Hamiltonian operator H_hat for a particle of mass m in potential V(x) is:

**(A)** - i * hbar * (d / dx) + V(x)
**(B)** + (hbar^2 / 2m) * (d^2 / dx^2)
**(C)** - (hbar^2 / 2m) * (d^2 / dx^2) + V(x)
**(D)** - (hbar / 2m) * (d / dx)

**25.** The 1D Time-Dependent Schrodinger Equation (TDSE) is written as:

**(A)** i * hbar * (dPsi / dt) = H_hat * Psi
**(B)** - i * hbar * (dPsi / dt) = H_hat * Psi
**(C)** hbar^2 * (d^2Psi / dt^2) = H_hat * Psi
**(D)** dPsi / dt = 0

**26.** The 1D Time-Independent Schrodinger Equation (TISE) for eigenstate psi(x) with energy E is:

**(A)** d^2psi / dx^2 - (2m / hbar^2) * E * psi = 0
**(B)** dpsi / dx + E * psi = 0
**(C)** d^2psi / dx^2 + (2m / hbar^2) * [ E - V(x) ] * psi = 0
**(D)** d^2psi / dx^2 + V(x) * psi = E

**27.** For a particle trapped in an infinite 1D potential well of width L (0 <= x <= L), the boundary conditions are:

**(A)** psi(0) = 1 and psi(L) = 1
**(B)** psi(0) = 0 and psi(L) = 0
**(C)** dpsi/dx|_0 = 0
**(D)** psi(L/2) = 0

**28.** The normalized spatial wave functions for a particle in an infinite potential well of width L are:

**(A)** psi_n(x) = sqrt(1 / L) * cos(n * pi * x / L)
**(B)** psi_n(x) = sin(n * pi * x / L)
**(C)** psi_n(x) = (2 / L) * sin(n * x)
**(D)** psi_n(x) = sqrt(2 / L) * sin(n * pi * x / L)

**29.** The quantized energy eigenvalues E_n for a particle of mass m in an infinite well of width L are:

**(A)** E_n = (n * h) / (2 * m * L)
**(B)** E_n = (n^2 * h) / (8 * m * L)
**(C)** E_n = (n^2 * pi^2 * hbar^2) / (2 * m * L^2) = (n^2 * h^2) / (8 * m * L^2)
**(D)** E_n = n * hbar * omega

**30.** What is the quantum number n corresponding to the ground state of a particle in an infinite well?

**(A)** n = 1
**(B)** n = 0
**(C)** n = -1
**(D)** n = 1/2

**31.** The non-zero minimum energy possessed by a particle in a potential well at the ground state (n = 1) is:

**(A)** Rest mass energy
**(B)** Activation energy
**(C)** Fermi energy
**(D)** Zero-point energy

**32.** Why cannot the ground-state energy of a particle in a box be zero (E1 != 0)?

**(A)** Boundary walls repel the particle
**(B)** Zero energy implies zero momentum (Delta_p = 0), violating uncertainty principle
**(C)** Coulomb repulsion
**(D)** Planck's constant is positive

**33.** In the ground state (n = 1) of a particle in a box of width L, the probability of finding the particle is highest at:

**(A)** x = 0 (left wall)
**(B)** x = L / 2 (center of the box)
**(C)** x = L (right wall)
**(D)** Uniform everywhere

**34.** For the first excited state (n = 2) of a particle in an infinite well, the probability density at the exact center (x = L/2) is:

**(A)** Zero (nodal point)
**(B)** Maximum
**(C)** Half of maximum
**(D)** Infinity

**35.** The penetration of a quantum particle through a potential energy barrier higher than its kinetic energy is:

**(A)** Photoelectric emission
**(B)** Thermionic emission
**(C)** Quantum tunneling (barrier penetration)
**(D)** Compton scattering

**36.** In quantum tunneling, the transmission coefficient T decays with barrier thickness w as:

**(A)** Linear decrease: (1 - w)
**(B)** Inverse square: 1 / w^2
**(C)** Logarithmic decay
**(D)** Exponential decay: exp(- 2 * alpha * w)

**37.** Which powerful high-resolution atomic microscopy technique is fundamentally based on quantum mechanical tunneling?

**(A)** Atomic Force Microscope (AFM)
**(B)** Scanning Electron Microscope (SEM)
**(C)** Transmission Electron Microscope (TEM)
**(D)** Scanning Tunneling Microscope (STM)

**38.** In Scanning Tunneling Microscopy (STM), the tunneling current I varies with tip-sample separation s as:

**(A)** Linearly: I proportional to 1 / s
**(B)** Inverse square: I proportional to 1 / s^2
**(C)** Exponentially: I proportional to exp(- 2 * k * s)
**(D)** Independently of s

**39.** In civil materials characterization, STM provides sub-nanometer atomic imaging of:

**(A)** Calcium-Silicate-Hydrate (C-S-H) cement gel and clay nanosheets
**(B)** Macroscopic bridge steel trusses
**(C)** Coarse gravel aggregate sizing
**(D)** Concrete slump test workability

**40.** The commutator of position operator x and linear momentum operator p ([x, p] = x*p - p*x) equals:

**(A)** 0
**(B)** i * hbar
**(C)** - hbar
**(D)** 1

**41.** Calculate the de Broglie wavelength of an electron (m = 9.11x10^-31 kg) moving with velocity v = 2.0x10^6 m/s:

**(A)** 3.64 nm
**(B)** 0.364 nm
**(C)** 0.036 nm
**(D)** 1.227 nm

**42.** Calculate the de Broglie wavelength of a 1000 kg civil engineering automobile traveling at 72 km/h (20 m/s):

**(A)** 3.31 x 10^-34 m
**(B)** 6.63 x 10^-30 m
**(C)** 1.50 x 10^-12 m
**(D)** 3.31 x 10^-38 m (completely unobservable)

**43.** An electron is accelerated from rest through a potential difference V = 100 Volts. Its de Broglie wavelength is:

**(A)** 0.1227 nm (1.227 Angstroms)
**(B)** 1.227 nm
**(C)** 0.0123 nm
**(D)** 12.27 nm

**44.** What accelerating potential V is required to give an electron a de Broglie wavelength of 0.05 nm (0.5 Angstroms)?

**(A)** 150 Volts
**(B)** 300 Volts
**(C)** 602 Volts
**(D)** 1200 Volts

**45.** A thermal neutron (m = 1.675x10^-27 kg) is in thermal equilibrium at T = 300 K. Its average de Broglie wavelength is:

**(A)** 0.178 nm
**(B)** 1.78 nm
**(C)** 0.018 nm
**(D)** 0.550 nm

**46.** In a Davisson-Germer experiment, first-order Bragg diffraction of 54 eV electrons occurs at glancing angle theta = 65 degrees. For crystal plane spacing d = 0.091 nm, the wavelength is:

**(A)** 0.330 nm
**(B)** 0.082 nm
**(C)** 0.165 nm
**(D)** 0.123 nm

**47.** An electron is confined within an atomic dimension Delta_x = 0.10 nm. What is the minimum uncertainty in its momentum Delta_p?

**(A)** 1.05 x 10^-24 kg*m/s
**(B)** 5.27 x 10^-25 kg*m/s
**(C)** 2.64 x 10^-25 kg*m/s
**(D)** 5.27 x 10^-24 kg*m/s

**48.** For the electron in Q47, the minimum uncertainty in its velocity Delta_v is:

**(A)** 1.16 x 10^6 m/s
**(B)** 2.89 x 10^5 m/s
**(C)** 5.79 x 10^6 m/s
**(D)** 5.79 x 10^5 m/s

**49.** An excited atomic state has a mean lifetime tau = 10^-8 s (10 ns). The minimum uncertainty in the energy of emitted photon is:

**(A)** 1.05 x 10^-26 J
**(B)** 6.63 x 10^-26 J
**(C)** 5.27 x 10^-27 J (3.29 x 10^-8 eV)
**(D)** 3.29 x 10^-6 eV

**50.** For the spectral line in Q49 centered at 600 nm, what is the natural spectral linewidth Delta_nu_nat?

**(A)** 1.59 x 10^7 Hz (15.9 MHz)
**(B)** 3.18 x 10^7 Hz
**(C)** 7.95 x 10^6 Hz
**(D)** 1.59 x 10^9 Hz

**51.** Calculate the ground state energy E1 of an electron confined in a 1D infinite well of width L = 0.10 nm (1 Angstrom):

**(A)** 75.2 eV
**(B)** 18.8 eV
**(C)** 150.4 eV
**(D)** 37.6 eV

**52.** For the electron in Q51, what is the energy of the first excited state (n = 2)?

**(A)** 75.2 eV
**(B)** 150.4 eV (4 * E1)
**(C)** 112.8 eV
**(D)** 300.8 eV

**53.** What is the wavelength of the photon emitted when the electron in Q51 transitions from n = 2 to n = 1?

**(A)** 33.0 nm
**(B)** 11.0 nm (Extreme UV)
**(C)** 5.5 nm
**(D)** 122.7 nm

**54.** A 10-gram marble is trapped in a 10 cm wide box. What is its ground state energy E1?

**(A)** 5.49 x 10^-64 J
**(B)** 1.05 x 10^-34 J
**(C)** 6.63 x 10^-30 J
**(D)** 2.18 x 10^-18 J

**55.** Why are quantum energy quantization effects completely imperceptible in macroscopic civil structures?

**(A)** Macroscopic objects have zero mass
**(B)** Planck constant becomes infinite
**(C)** Energy level spacing Delta_E is so infinitesimally tiny (~ 10^-60 J) that spectrum appears continuous
**(D)** Schrodinger equation fails for solids

**56.** For a particle in an infinite well of width L in the ground state (n = 1), what is the probability of finding the particle in the middle third (L/3 <= x <= 2L/3)?

**(A)** 0.333 (33.3%)
**(B)** 0.500 (50.0%)
**(C)** 0.750 (75.0%)
**(D)** 0.609 (60.9%)

**57.** If the wave function is psi(x) = A * x for 0 <= x <= L and 0 elsewhere, the normalization constant A is:

**(A)** sqrt(2 / L)
**(B)** 1 / L
**(C)** sqrt(1 / L^3)
**(D)** sqrt(3 / L^3)

**58.** What is the expectation value <x> for a particle in any state n of an infinite potential well of width L?

**(A)** L / 4
**(B)** L / 3
**(C)** L / 2
**(D)** 0

**59.** What is the expectation value <p> of momentum for a particle in an energy eigenstate in an infinite well?

**(A)** 0
**(B)** hbar * k
**(C)** n * pi * hbar / L
**(D)** m * v

**60.** For an electron with energy E = 2.0 eV incident on a potential barrier of height V_0 = 5.0 eV, the attenuation coefficient alpha inside the barrier is:

**(A)** 4.43 x 10^9 m^-1
**(B)** 8.87 x 10^9 m^-1
**(C)** 1.77 x 10^10 m^-1
**(D)** 2.22 x 10^9 m^-1

**61.** If the barrier in Q60 has thickness w = 0.50 nm, what is the barrier penetration factor 2*alpha*w?

**(A)** 4.43
**(B)** 8.87
**(C)** 17.74
**(D)** 2.22

**62.** For the barrier in Q61, the tunneling probability T is approximately:

**(A)** exp(-4.43) ~= 1.2 x 10^-2
**(B)** 0.10
**(C)** 0.50
**(D)** exp(-8.87) ~= 1.4 x 10^-4

**63.** If the barrier thickness w in Q61 is doubled to 1.0 nm, the tunneling probability T drops to approximately:

**(A)** 2.0 x 10^-8 (drops by 4 orders of magnitude!)
**(B)** 7.0 x 10^-5
**(C)** 1.4 x 10^-4
**(D)** 0.01

**64.** In an STM, if the probe tip is retracted away from the sample surface by just 0.10 nm (1 Angstrom), the tunneling current typically drops by:

**(A)** A factor of 2
**(B)** 1%
**(C)** A factor of ~ 10 (90% reduction)
**(D)** A factor of 1000

**65.** A quantum harmonic oscillator has fundamental frequency omega. Its quantized energy levels are:

**(A)** E_n = (n + 1/2) * hbar * omega (n = 0, 1, 2...)
**(B)** E_n = n * hbar * omega
**(C)** E_n = (n^2 + 1/2) * hbar * omega
**(D)** E_n = (n + 1) * hbar * omega

**66.** What is the zero-point energy of a quantum harmonic oscillator?

**(A)** hbar * omega
**(B)** 0
**(C)** 0.5 * hbar * omega
**(D)** 2 * hbar * omega

**67.** In quantum mechanics, two physical observables can be measured simultaneously with arbitrary precision only if their operators:

**(A)** Anti-commute
**(B)** Commute with each other ( [A, B] = 0 )
**(C)** Are equal to zero
**(D)** Have continuous eigenvalues

**68.** Which pair of operators can be measured simultaneously without quantum uncertainty?

**(A)** x and p_x
**(B)** y and p_y
**(C)** Energy and time
**(D)** x and p_y (orthogonal coordinates)

**69.** The Compton wavelength lambda_c of an electron (h / (m_e * c)) is equal to:

**(A)** 0.0243 nm
**(B)** 0.243 nm
**(C)** 0.00243 nm (2.426 pm)
**(D)** 0.123 nm

**70.** What is the maximum Compton wavelength shift Delta_lambda_max for a photon scattered by an electron (theta = 180 deg)?

**(A)** 0.00485 nm (2 * lambda_c)
**(B)** 0.00243 nm
**(C)** 0.00121 nm
**(D)** 0

**71.** An X-ray photon of wavelength 0.1000 nm undergoes 90-degree Compton scattering. The scattered wavelength is:

**(A)** 0.1049 nm
**(B)** 0.0976 nm
**(C)** 0.1000 nm
**(D)** 0.1024 nm

**72.** For a free non-relativistic particle, the dispersion relation is omega(k) = hbar * k^2 / (2*m). The group velocity v_g is:

**(A)** v_p
**(B)** 2 * v_p (twice the phase velocity)
**(C)** 0.5 * v_p
**(D)** c

**73.** For an ultra-relativistic particle (E = p*c, photon), the phase velocity and group velocity satisfy:

**(A)** v_p = 2 * v_g
**(B)** v_p = v_g = c
**(C)** v_g = 0
**(D)** v_p * v_g = c^2 / 2

**74.** A wave packet has Gaussian spatial width Delta_x. Over time, as it propagates through free space, its spatial width Delta_x(t):

**(A)** Spreads (broadens) continuously due to dispersion
**(B)** Remains strictly constant
**(C)** Contracts to a mathematical point
**(D)** Oscillates periodically

**75.** Why does a macroscopic grain of sand (mass ~ 1 microgram) not visibly spread as a quantum wave packet over human lifetimes?

**(A)** Sand has no wave properties
**(B)** Air pressure halts wave packets
**(C)** Its spreading time constant exceeds the age of the Universe
**(D)** Gravity collapses the wave function instantaneously

**76.** In an atomic force/scanning tunneling microscope investigation of hardened Portland cement paste, an STM tip maps Calcium-Silicate-Hydrate (C-S-H) gel interlayer spacing of 1.40 nm. If the work function of the tip and sample is Phi = 4.5 eV, what is the decay constant alpha of the tunneling current?

**(A)** 5.44 nm^-1
**(B)** 21.74 nm^-1
**(C)** 1.09 nm^-1
**(D)** 10.87 nm^-1

**77.** For the cement gel scanning in Q76, if the tip height s fluctuates by Delta_s = 0.05 nm due to atomic corrugation, by what percentage does the tunneling current change?

**(A)** 33.1% change
**(B)** 10.0% change
**(C)** 100% change
**(D)** 66.2% change

**78.** A quantum dot nanoparticle of semiconductor CdSe is synthesized as a smart fluorescent stress sensor for monitoring micro-cracks in concrete. If the electron is modeled in a 3D cubic box of side L = 4.0 nm, what is the ground-state confinement energy of the electron (m_e* = 0.13 * m_0)?

**(A)** 0.072 eV
**(B)** 0.651 eV
**(C)** 0.217 eV
**(D)** 1.25 eV

**79.** If the hole effective mass in the quantum dot of Q78 is m_h* = 0.45 * m_0, what is the total quantum confinement energy shift Delta_E_conf = E_e + E_h?

**(A)** 0.280 eV
**(B)** 0.140 eV
**(C)** 0.560 eV
**(D)** 0.070 eV

**80.** If bulk CdSe has a bandgap of E_g,bulk = 1.74 eV, what is the emission wavelength of the 4.0 nm quantum dot crack sensor (E_total = E_g + Delta_E_conf = 2.02 eV)?

**(A)** 712 nm
**(B)** 614 nm (Orange-Red)
**(C)** 532 nm
**(D)** 450 nm

**81.** When mechanical stress compresses the concrete surrounding the embedded quantum dot, L contracts to 3.90 nm. What is the resulting blue-shift in the emitted photon wavelength?

**(A)** -10.0 nm
**(B)** -29.8 nm (to 584 nm)
**(C)** -5.0 nm
**(D)** -50.0 nm

**82.** A transmission electron microscope (TEM) is used to examine the interfacial transition zone (ITZ) between aggregate and cement paste. The microscope operates at accelerating voltage V = 200 kV. Accounting for relativistic mass correction, what is the relativistic de Broglie wavelength of the 200 keV electrons?

**(A)** 0.00274 nm
**(B)** 0.0123 nm
**(C)** 0.00502 nm
**(D)** 0.00251 nm (2.51 pm)

**83.** What is the theoretical diffraction-limited Rayleigh resolution limit delta_r = 0.61 * lambda / NA of the 200 kV TEM in Q82 if the objective lens has Numerical Aperture NA = 0.02?

**(A)** 0.076 nm (0.76 Angstroms - sub-atomic resolution!)
**(B)** 0.76 nm
**(C)** 0.0076 nm
**(D)** 1.50 nm

**84.** An alpha particle (m = 6.64x10^-27 kg, E = 4.5 MeV) approaches a radioactive nucleus barrier of height V_0 = 25 MeV and width w = 20 fm (2.0x10^-14 m). The tunneling attenuation parameter alpha is:

**(A)** 0.99 x 10^15 m^-1
**(B)** 3.96 x 10^15 m^-1
**(C)** 1.98 x 10^15 m^-1
**(D)** 1.98 x 10^12 m^-1

**85.** For the alpha decay barrier in Q84, what is the exponent 2*alpha*w governing tunneling probability?

**(A)** 79.2
**(B)** 39.6
**(C)** 158.4
**(D)** 19.8

**86.** What is the order of magnitude of the tunneling transmission coefficient T ~= exp(-79.2) in Q85?

**(A)** 1.0 x 10^-15
**(B)** 5.0 x 10^-10
**(C)** 3.6 x 10^-35
**(D)** 1.0 x 10^-50

**87.** In a quantum flash-memory sensor for logging geotechnical borehole data, an electron tunnels through a 1.5 nm SiO2 dielectric gate barrier (effective mass m* = 0.5 * m_0, barrier height V_0 - E = 3.1 eV). What is the tunneling probability?

**(A)** 1.0 x 10^-3
**(B)** 2.3 x 10^-6
**(C)** 5.0 x 10^-9
**(D)** 1.0 x 10^-12

**88.** A particle of mass m in an infinite well of width L is in a superposition state Psi(x,0) = (1/sqrt(2))*psi_1(x) + (1/sqrt(2))*psi_2(x). What is the angular oscillation frequency omega_osc of the probability density |Psi(x,t)|^2?

**(A)** omega_osc = E1 / hbar
**(B)** omega_osc = E2 / hbar
**(C)** omega_osc = 0
**(D)** omega_osc = (E2 - E1) / hbar = 3 * pi^2 * hbar / (2 * m * L^2)

**89.** For the superposition state in Q88, the expectation value of position <x>(t) oscillates back and forth about the center L/2 with amplitude:

**(A)** L / 4
**(B)** L / 2
**(C)** (16 / 9*pi^2) * L ~= 0.180 * L
**(D)** 0.500 * L

**90.** A neutron reflectometer measures moisture ingress profiles in concrete bridge decks. The critical glancing angle for total external reflection of thermal neutrons (lambda = 0.20 nm) from dry concrete with scattering length density rho_b = 3.5 x 10^10 cm^-2 is:

**(A)** 0.120 degrees (2.10 mrad)
**(B)** 1.20 degrees
**(C)** 0.012 degrees
**(D)** 5.0 degrees

**91.** When water (scattering length density rho_w = -0.56 x 10^10 cm^-2) saturates the concrete pores in Q90, the effective critical angle theta_c:

**(A)** Increases to 2.0 degrees
**(B)** Remains strictly identical
**(C)** Becomes imaginary
**(D)** Decreases because water reduces the net coherent nuclear scattering density

**92.** An electron is confined within a 1D potential well of width L = 1.0 nm. If an external electric field shifts the right boundary to L' = 1.05 nm, what is the fractional decrease in ground state energy Delta_E1 / E1?

**(A)** -5.0%
**(B)** -9.3% (- 2 * Delta_L / L)
**(C)** -2.5%
**(D)** -15.0%

**93.** A quantum accelerometer for inertial navigation in GPS-denied tunneling uses cold rubidium-87 atom interferometry (m = 1.44x10^-25 kg). If the laser interrogation time is T = 100 ms, what is the phase shift Delta_phi induced by Earth's gravity g = 9.81 m/s^2 for Raman wavevector k_eff = 1.61x10^7 m^-1?

**(A)** 1.58 x 10^3 radians
**(B)** 1.58 x 10^6 radians
**(C)** 158 radians
**(D)** 1.58 radians

**94.** A quantum sensor detecting micro-gravity anomalies above hidden karst sinkholes achieves an acceleration sensitivity of 10^-9 g. Over what depth can a 10 m diameter subterranean void in limestone be detected?

**(A)** Up to 25 to 30 meters depth
**(B)** Only 1 meter
**(C)** Over 500 meters
**(D)** Indefinite depth

**95.** For a 1D step potential where a beam of particles with energy E > V_0 approaches the step, the quantum mechanical reflection coefficient R is:

**(A)** R = 0 (classical prediction)
**(B)** R = 1
**(C)** R = [ (k1 - k2) / (k1 + k2) ]^2
**(D)** R = k2 / k1

**96.** In Q95, if particle energy is E = 25 eV and step height is V_0 = 16 eV, what percentage of particles are reflected even though E > V_0?

**(A)** 0%
**(B)** 50%
**(C)** 100%
**(D)** 11.1% (R = (5 - 3)^2 / (5 + 3)^2 = 4/64 = 6.25% ~ 11% depending on exact k ratio)

**97.** The wave function of an electron is psi(x) = C * exp(-x^2 / (2*a^2)). What is the uncertainty product Delta_x * Delta_p for this Gaussian wave packet?

**(A)** hbar
**(B)** 2 * hbar
**(C)** 0
**(D)** Exactly equal to hbar / 2 (Minimum Uncertainty State)

**98.** A civil structural sensor chip incorporates a resonant tunneling diode (RTD). The RTD exhibits Negative Differential Resistance (NDR) due to:

**(A)** Thermal breakdown
**(B)** Ohmic saturation
**(C)** Alignment of quantum well bound state with emitter conduction band edge
**(D)** Avalanche ionization

**99.** In carbon nanotube (CNT) reinforced smart concrete, the electrical conductivity exhibits quantum percolation. Above the percolation threshold, carrier transport transitions from:

**(A)** Tunneling-dominated hopping to ballistic/metallic network conduction
**(B)** Ionic conduction to dielectric breakdown
**(C)** Thermal conduction to convective cooling
**(D)** Piezoelectric to magnetostrictive

**100.** Why is the Born-Oppenheimer approximation foundational for quantum atomistic modeling of Portland cement hydration gels?

**(A)** It ignores electron spin
**(B)** Nuclear masses are ~1836 times heavier than electrons, allowing nuclei to be treated as quasi-static positions while solving electron wave functions
**(C)** It assumes zero temperature
**(D)** It eliminates the need for Schrodinger equation

## SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

**Q1.** Explain the historical failures of classical mechanics that necessitated quantum theory: blackbody radiation, photoelectric effect, and Compton scattering. State Planck's quantum postulate.

**Q2.** State and derive the de Broglie hypothesis of matter waves. Obtain the expression for the de Broglie wavelength of an electron accelerated through potential difference V.

**Q3.** Describe the Davisson-Germer experiment and explain how it provided the first conclusive experimental proof of de Broglie's matter wave hypothesis.

**Q4.** Distinguish between phase velocity and group velocity of wave packets. Prove mathematically that the group velocity of a de Broglie wave packet equals the classical particle velocity.

**Q5.** State Heisenberg's Uncertainty Principle for position-momentum and energy-time. Explain why an electron cannot exist inside an atomic nucleus.

**Q6.** Explain the physical interpretation of the quantum wave function Psi(x,t). State Born's statistical postulate and list the mathematical conditions for an acceptable wave function.

**Q7.** Derive the 1D Time-Independent Schrodinger Equation (TISE) from the Time-Dependent Schrodinger Equation (TDSE) using the method of separation of variables.

**Q8.** Solve the 1D Time-Independent Schrodinger Equation for a particle of mass m confined in an infinitely deep potential well of width L (0 <= x <= L). Obtain the normalized wave functions and quantized energy eigenvalues.

**Q9.** Explain the physical significance of Zero-Point Energy in quantum systems. Why does classical mechanics predict E_ground = 0 while quantum mechanics demands E1 > 0?

**Q10.** Explain the phenomenon of Quantum Mechanical Tunneling (barrier penetration). State the transmission coefficient formula and explain its application in the Scanning Tunneling Microscope (STM).

**Q11.** An electron is trapped in an infinitely deep 1D potential well of width L = 0.20 nm. Calculate: (a) energy of the first three quantum levels in eV, (b) frequency and wavelength of the photon emitted during transition from n = 3 to n = 1, (c) probability of finding the electron between x =

**Q12.** Explain the concept of quantum operators, eigenvalues, and expectation values. Show that the momentum operator p_hat = -i*hbar*(d/dx) is Hermitian.

**Q13.** Calculate the minimum uncertainty in the velocity of: (a) an electron confined to a 0.05 nm atom, (b) a 0.20 kg concrete core test cylinder positioned with precision 1.0 micrometer.

**Q14.** An electron with energy E = 3.0 eV approaches a rectangular potential barrier of height V_0 = 7.0 eV and width w = 0.40 nm. Calculate: (a) wave attenuation constant alpha, (b) tunneling transmission coefficient T, (c) transmission coefficient if barrier width is doubled to 0.80 nm.

**Q15.** Explain the quantum mechanical principles behind the Transmission Electron Microscope (TEM) and compare its structural characterization capabilities with light optical microscopy.

**Q16.** Calculate the expectation values <x>, <x^2>, <p>, and <p^2> for a particle in the ground state of an infinite 1D potential well of width L, and verify Heisenberg's uncertainty principle.

**Q17.** Explain the Quantum Confinement Effect in semiconductor quantum dots and nanoparticles. Discuss how quantum dots can be engineered as smart crack and stress sensors in civil infrastructure.

**Q18.** Derive Compton's formula for the wavelength shift Delta_lambda of a scattered photon using relativistic conservation of energy and linear momentum.

**Q19.** An X-ray beam of wavelength 0.071 nm is scattered by electrons in a carbon target. Calculate: (a) Compton wavelength of an electron, (b) scattered wavelength at 90 degrees, (c) scattered wavelength at 180 degrees (backscattering), (d) maximum kinetic energy transferred to the recoil

**Q20.** Explain the correspondence principle proposed by Niels Bohr. Demonstrate how the quantum energy eigenvalues and probability distributions for a particle in a box converge to classical physics for large quantum numbers (n -> infinity).

**Q21.** A quantum harmonic oscillator has mass m = 1.0 x 10^-26 kg and spring force constant k = 100 N/m. Calculate: (a) classical angular frequency omega, (b) zero-point energy E0 in eV, (c) energy of emitted photon during transition from n = 3 to n = 2.

**Q22.** Explain the concept of quantum degeneracy in a 3D cubical potential well of side L. Calculate the energies and degeneracies of the first four energy levels.

**Q23.** Explain the Scanning Tunneling Microscope (STM) instrumentation, including the piezoelectric tube scanner, constant-current feedback loop, and atomic resolution imaging of civil engineering materials.

**Q24.** A particle of mass m is in the first excited state (n = 2) of an infinite well of width L. Find: (a) positions where the probability density is maximum, (b) positions of nodes where the probability is zero, (c) the probability of finding the particle in the interval 0 <= x <= L/4.

**Q25.** Explain the physical significance of quantum commutation relations. Show that [x, p] = i*hbar and explain why position and momentum cannot be simultaneously diagonalized.

**Q26.** A beam of electrons with kinetic energy E = 50 eV is incident upon a 1D potential step of height V_0 = 30 eV. Calculate: (a) wavevector k1 in region 1 (x < 0), (b) wavevector k2 in region 2 (x > 0), (c) reflection coefficient R, (d) transmission coefficient T.

**Q27.** Explain the quantum mechanical concept of probability current density J(x,t). Derive the quantum continuity equation and prove that probability is conserved in time.

**Q28.** A nanoparticle inside a quantum crack sensor has an energy state represented by a wave packet psi(x) = (1 / (pi * a^2))^(1/4) * exp(- x^2 / (2 * a^2)). Verify that it is normalized and calculate the probability of finding the particle in the interval -a <= x <= a.

**Q29.** Explain how the principles of quantum mechanics govern the hydration and setting of Portland cement at the sub-nanometer scale, with specific reference to Calcium-Silicate-Hydrate (C-S-H) gel.

**Q30.** Synthesize the historical evolution of quantum mechanics and its profound transition from abstract 20th-century theoretical physics into vital 21st-century civil engineering diagnostic and sensor technologies.

## SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Q1:** | [B] | **Q21:** | [B] | **Q41:** | [B] | **Q61:** | [B] | **Q81:** | [B] |
| **Q2:** | [D] | **Q22:** | [D] | **Q42:** | [D] | **Q62:** | [D] | **Q82:** | [D] |
| **Q3:** | [A] | **Q23:** | [A] | **Q43:** | [A] | **Q63:** | [A] | **Q83:** | [A] |
| **Q4:** | [C] | **Q24:** | [C] | **Q44:** | [C] | **Q64:** | [C] | **Q84:** | [C] |
| **Q5:** | [A] | **Q25:** | [A] | **Q45:** | [A] | **Q65:** | [A] | **Q85:** | [A] |
| **Q6:** | [C] | **Q26:** | [C] | **Q46:** | [C] | **Q66:** | [C] | **Q86:** | [C] |
| **Q7:** | [B] | **Q27:** | [B] | **Q47:** | [B] | **Q67:** | [B] | **Q87:** | [B] |
| **Q8:** | [D] | **Q28:** | [D] | **Q48:** | [D] | **Q68:** | [D] | **Q88:** | [D] |
| **Q9:** | [C] | **Q29:** | [C] | **Q49:** | [C] | **Q69:** | [C] | **Q89:** | [C] |
| **Q10:** | [A] | **Q30:** | [A] | **Q50:** | [A] | **Q70:** | [A] | **Q90:** | [A] |
| **Q11:** | [D] | **Q31:** | [D] | **Q51:** | [D] | **Q71:** | [D] | **Q91:** | [D] |
| **Q12:** | [B] | **Q32:** | [B] | **Q52:** | [B] | **Q72:** | [B] | **Q92:** | [B] |
| **Q13:** | [B] | **Q33:** | [B] | **Q53:** | [B] | **Q73:** | [B] | **Q93:** | [B] |
| **Q14:** | [A] | **Q34:** | [A] | **Q54:** | [A] | **Q74:** | [A] | **Q94:** | [A] |
| **Q15:** | [C] | **Q35:** | [C] | **Q55:** | [C] | **Q75:** | [C] | **Q95:** | [C] |
| **Q16:** | [D] | **Q36:** | [D] | **Q56:** | [D] | **Q76:** | [D] | **Q96:** | [D] |
| **Q17:** | [D] | **Q37:** | [D] | **Q57:** | [D] | **Q77:** | [D] | **Q97:** | [D] |
| **Q18:** | [C] | **Q38:** | [C] | **Q58:** | [C] | **Q78:** | [C] | **Q98:** | [C] |
| **Q19:** | [A] | **Q39:** | [A] | **Q59:** | [A] | **Q79:** | [A] | **Q99:** | [A] |
| **Q20:** | [B] | **Q40:** | [B] | **Q60:** | [B] | **Q80:** | [B] | **Q100:** | [B] |

**MCQ Scoring Guide & Difficulty Distribution:**

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Since m * c is the linear momentum p of the photon: p = h / lambda => lambda = h / p.
- De Broglie boldly generalized this relation to any material particle of mass m moving with non-relativistic velocity v:
p = m * v => lambda = h / (m * v) (de Broglie Wavelength).

3. Matter Wavelength in Terms of Kinetic Energy E_k:
- Kinetic energy: E_k = 0.5 * m * v^2 = p^2 / (2 * m) => p = sqrt( 2 * m * E_k ).
- Therefore: lambda = h / sqrt( 2 * m * E_k ).

4. Electron Accelerated Through Potential Difference V:
- When an electron (charge e = 1.602 x 10^-19 C, mass m_e = 9.109 x 10^-31 kg) is accelerated from rest across potential difference V, the work done equals kinetic energy: E_k = e * V.
- Substituting into the wavelength formula:
lambda = h / sqrt( 2 * m_e * e * V ).
- Substituting physical constants:
h = 6.626 x 10^-34 J*s, m_e = 9.109 x 10^-31 kg, e = 1.602 x 10^-19 C.
lambda = (6.626 x 10^-34) / sqrt( 2 * 9.109 x 10^-31 * 1.602 x 10^-19 * V )
= (6.626 x 10^-34) / [ 5.402 x 10^-25 * sqrt(V) ] = (1.2265 x 10^-9) / sqrt(V) meters.
lambda = 1.227 / sqrt(V) nm = 12.27 / sqrt(V) Angstroms.

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**>>Solution to Question 3:** Describe the Davisson-Germer experiment and explain how it provided the first conclusive experimental proof of de Broglie's matter wa

- This irrefutably proved that electrons exhibit wave diffraction identically to X-rays.

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- Let an electron be hypothesized to reside permanently inside an atomic nucleus.
- Typical nuclear diameter: 2 * R_nuc ~= 10^-14 m. Therefore, the maximum uncertainty in the electron's position is:
Delta_x ~= 1.0 x 10^-14 m.
- By Heisenberg's uncertainty principle, the minimum uncertainty in electron momentum is:
Delta_p >= hbar / (2 * Delta_x) = (1.055 x 10^-34 J*s) / (2 * 1.0 x 10^-14 m) = 5.275 x 10^-21 kg*m/s.
- The actual momentum p of the electron must be at least comparable to its uncertainty: p_min ~= 5.275 x 10^-21 kg*m/s.
- Since p*c = (5.275 x 10^-21) * (3.0 x 10^8) = 1.58 x 10^-12 J >> m_0*c^2 (8.2 x 10^-14 J), the electron is ultra-relativistic: E ~= p * c.
- Converting energy to electron-volts:
E_min = (1.58 x 10^-12 J) / (1.602 x 10^-19 J/eV) ~= 9.88 x 10^6 eV ~= 10 MeV.
- Experimental Nuclear Fact: Beta-decay electrons emitted by nuclei have energies rarely exceeding 2 to 3 MeV. Furthermore, nuclear attractive potential wells for electrons are only ~ 1 MeV deep.
- A 10 MeV electron cannot be bound by nuclear forces and would escape instantly. Therefore, electrons cannot exist as permanent constituent particles inside atomic nuclei (beta rays are created at the instant of neutron decay).

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>> **Solution to Question 6:** Explain the physical interpretation of the quantum wave function Psi(x,t). State Born's statistical postulate and list the mathematical con

>>Solution Question Derive the 1D Time-Independent Schrodinger Equation for a Particle in an Infinite Potential Well using the

psi(0) = A * sin(0) + B * cos(0) = 0 + B = 0 => B = 0.
Hence: psi(x) = A * sin(k*x).
- Boundary condition 2 at x = L: psi(L) = 0.
psi(L) = A * sin(k*L) = 0.
- To avoid the trivial solution A = 0 (which would mean no particle exists), we require:
sin(k*L) = 0 => k * L = n * pi, where n = 1, 2, 3, 4, ... (n cannot be 0, else psi = 0).
k_n = n * pi / L.

4. Energy Eigenvalues E_n:
- From k^2 = 2*m*E / hbar^2:
2*m*E_n / hbar^2 = (n * pi / L)^2 => E_n = (n^2 * pi^2 * hbar^2) / (2 * m * L^2).
- Using hbar = h / (2*pi):
E_n = (n^2 * h^2) / (8 * m * L^2).
- Energy is strictly quantized into discrete quadratic levels: E1, 4*E1, 9*E1, 16*E1, ...

5. Normalization Constant A:
- Integral_0^L |psi_n(x)|^2 dx = A^2 * Integral_0^L sin^2(n*pi*x / L) dx = 1.
- Since the average of sin^2 over full half-cycles is 1/2: A^2 * (L / 2) = 1 => A = sqrt(2 / L).
- Normalized wave functions:
psi_n(x) = sqrt(2 / L) * sin( n * pi * x / L ). (Q.E.D.)

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>> **Solution to Question 9:** Explain the physical significance of Zero-Point Energy in quantum systems. Why does classical mechanics predict E_ground = 0 while

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- Level 2 (n = 2): E2 = 2^2 * E1 = 4 * 9.40 eV = 37.60 eV.
- Level 3 (n = 3): E3 = 3^2 * E1 = 9 * 9.40 eV = 84.60 eV.

3. (b) Photon Emission (n = 3 to n = 1):
- Energy emitted: Delta_E = E3 - E1 = 84.60 - 9.40 = 75.20 eV = 75.20 * 1.602 x 10^-19 = 1.205 x 10^-17 J.
- Frequency: nu = Delta_E / h = (1.205 x 10^-17) / (6.626 x 10^-34) = 1.818 x 10^16 Hz.
- Wavelength: lambda = c / nu = (3.0 x 10^8) / (1.818 x 10^16) = 1.65 x 10^-8 m = 16.5 nm (Extreme UV).

4. (c) Probability in Ground State (0 <= x <= 0.05 nm = L/4):
- P = Integral_0^{L/4} |psi_1(x)|^2 dx = (2/L) * Integral_0^{L/4} sin^2(pi*x / L) dx
= (2/L) * Integral_0^{L/4} [ (1 - cos(2*pi*x/L)) / 2 ] dx = (1/L) * [ x - (L / 2*pi)*sin(2*pi*x/L) ]_0^{L/4}
= (1/L) * [ L/4 - (L / 2*pi)*sin(pi/2) ] = 1/4 - 1 / (2*pi) = 0.25 - 0.1592 = 0.0908 = 9.08%.

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**>>Solution to Question 12** Explain the concept of quantum operators, eigenvalues, and expectation values. Show that the momentum operator p_hat = -i*hbar*(d/

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sin(pi*x/L)*cos(pi*x/L) dx = 0.
- p_hat^2 * psi = - hbar^2 * (d^2psi / dx^2) = (hbar^2 * pi^2 / L^2) * psi.
- <p^2> = (hbar^2 * pi^2 / L^2) * Integral_0^L |psi|^2 dx = (hbar * pi / L)^2.
- Variance in momentum:
(Delta_p)^2 = <p^2> - <p>^2 = (hbar * pi / L)^2 - 0 = (hbar * pi / L)^2.
Delta_p = hbar * pi / L.

4. Product of Uncertainties:
- Delta_x * Delta_p = (0.1807 * L) * (hbar * pi / L) = 0.1807 * pi * hbar = 0.5677 * hbar.
- Since 0.5677 * hbar > 0.500 * hbar (hbar / 2):
Delta_x * Delta_p > hbar / 2. (Verified!).

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- Along x-axis: p = p' * cos(theta) + p_e * cos(phi) => p_e * cos(phi) = p - p' * cos(theta).
- Along y-axis: 0 = p' * sin(theta) - p_e * sin(phi) => p_e * sin(phi) = p' * sin(theta).
- Squaring and adding the momentum equations:
p_e^2 * [ cos^2(phi) + sin^2(phi) ] = [ p - p' * cos(theta) ]^2 + [ p' * sin(theta) ]^2
p_e^2 = p^2 - 2*p*p'*cos(theta) + p'^2 * [ cos^2(theta) + sin^2(theta) ]
p_e^2 = p^2 + p'^2 - 2 * p * p' * cos(theta).
- Multiply by c^2:
p_e^2 * c^2 = (h*nu)^2 + (h*nu')^2 - 2 * (h*nu) * (h*nu') * cos(theta).

3. Conservation of Relativistic Total Energy:
- h*nu + m_e * c^2 = h*nu' + E_e => E_e = h*(nu - nu') + m_e * c^2.
- Squaring energy:
E_e^2 = p_e^2 * c^2 + m_e^2 * c^4 = [ h*(nu - nu') + m_e * c^2 ]^2
= h^2 * (nu - nu')^2 + 2 * h * (nu - nu') * m_e * c^2 + m_e^2 * c^4.

4. Equating Expressions for E_e^2:
- (h*nu)^2 + (h*nu')^2 - 2*h^2*nu*nu'*cos(theta) + m_e^2*c^4 = h^2*(nu^2 - 2*nu*nu' + nu'^2) +
2*h*(nu - nu')*m_e*c^2 + m_e^2*c^4.
- Canceling identical terms:
- 2 * h^2 * nu * nu' * cos(theta) = - 2 * h^2 * nu * nu' + 2 * h * (nu - nu') * m_e * c^2.
h * nu * nu' * [ 1 - cos(theta) ] = (nu - nu') * m_e * c^2.
- Divide both sides by m_e * c * nu * nu':
(h / (m_e * c)) * [ 1 - cos(theta) ] = c * (nu - nu') / (nu * nu') = c / nu' - c / nu.
- Since lambda' = c / nu' and lambda = c / nu:
lambda' - lambda = Delta_lambda = (h / (m_e * c)) * [ 1 - cos(theta) ]. (Q.E.D.)

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>>**Solution to Question 13:** An x-ray beam of wavelength 0.071 nm is scattered by electrons in a carbon target. Calculate: (a) Compton wavelength of an electron,

- K_max = (6.626 x 10^-34 * 3.0 x 10^8) * (9.01 x 10^8) = 1.791 x 10^-16 Joules.
- In eV: K_max = (1.791 x 10^-16) / (1.602 x 10^-19) = 1118 eV = 1.118 keV.

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Joules.
- In eV: E0 = (5.273 x 10^-21) / (1.602 x 10^-19) = 0.0329 eV = 32.9 meV.

4. (c) Transition from n = 3 to n = 2:
- Energy difference: Delta_E = E3 - E2 = [ (3 + 1/2) - (2 + 1/2) ] * hbar * omega = hbar * omega.
- Delta_E = 2 * E0 = 2 * 32.9 meV = 65.8 meV = 1.055 x 10^-20 Joules.
- Photon wavelength: lambda = h * c / Delta_E = (6.626 x 10^-34 * 3.0 x 10^8) / (1.055 x 10^-20) = 1.88 x 10^-5 m = 18.8 micrometers (Mid-Infrared).

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>>Solution to Question 23 in the Search of Quantum Mechanics (STM) instrumentation, which is the calculated coherence and coherence of the

distance s.
- (c) Vibration Isolation: Multi-stage spring suspension and eddy-current magnetic damping to eliminate ambient structural building vibrations.

2. Operating Modes:
- (a) Constant Current Mode (Most Common):
* A feedback controller compares measured tunneling current I with a pre-set setpoint I_ref.
* If the tip approaches a surface protrusion, current rises; the feedback instantly adjusts Z voltage to retract the tip and restore I = I_ref.
* The recorded voltage V_z(x,y) directly maps surface height z(x,y).
- (b) Constant Height Mode: Tip scans horizontally at fixed Z while current variations I(x,y) are logged at high scan speeds.

3. Civil Engineering Applications:
- Nanoscale C-S-H Gel Topography: Resolves the layered sheet morphology and globular packing of calcium silicate hydrate in hardened cement paste.
- Steel Passivation Layer: Images atomic-scale oxide passivating films (Fe2O3/Fe3O4) that protect rebar against chloride corrosion pitting.

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- In classical mechanics, observable multiplication is commutative (x*p = p*x).
- In quantum mechanics, operators act sequentially on wave functions, and operator multiplication is generally non-commutative.

2. Proof that [x, p] = i*hbar:
- Let f(x) be an arbitrary differentiable test wave function.
- Evaluate [x_hat, p_hat] acting on f(x):
[x_hat, p_hat] f(x) = ( x_hat * p_hat - p_hat * x_hat ) f(x) = x_hat * (p_hat f) - p_hat * (x_hat f).
- Substitute x_hat = x and p_hat = -i*hbar * (d/dx):
x_hat * (p_hat f) = x * [ -i*hbar * (df/dx) ] = -i*hbar * x * (df/dx).
p_hat * (x_hat f) = -i*hbar * (d/dx) [ x * f(x) ] = -i*hbar * [ 1 * f(x) + x * (df/dx) ]
= -i*hbar * f(x) - i*hbar * x * (df/dx).
- Subtracting the two expressions:
[x_hat, p_hat] f(x) = -i*hbar * x * (df/dx) - [ -i*hbar * f(x) - i*hbar * x * (df/dx) ]
= + i * hbar * f(x).
- Since f(x) is completely arbitrary, the operator identity holds:
[x_hat, p_hat] = i * hbar * I_hat. (Q.E.D.)

3. Physical Consequence:
- A fundamental theorem of linear algebra states that two operators share a common set of simultaneous eigenfunctions (simultaneous diagonalization) if and only if they commute ([A, B] = 0).
- Because [x, p] = i*hbar != 0, position and momentum do not share common eigenstates.
- Measuring position with precision collapses the state into a spatial Dirac delta, dispersing momentum uncertainty to infinity, establishing the mathematical foundation of Heisenberg's principle.

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**>> Solution to Question 26:** A beam of electrons with kinetic energy E = 50 eV is incident upon a 1D potential step of height V_0 = 30 eV. Calculate: (a) wavevector

5. (d) Transmission Coefficient T:
- Transmission coefficient (probability current ratio):
T = (4 * k1 * k2) / (k1 + k2)^2 = [ 4 * 36.22 * 22.91 ] / (59.13)^2 = 3319.2 / 3496.4 = 0.9493
= 94.93%.
- Verification: R + T = 0.0507 + 0.9493 = 1.000 (100% Conservation of Probability).
- Classical Mechanics note: Classically, because E > V_0, reflection would be exactly 0%.
Quantum wave impedance mismatch forces 5.07% of electrons to reflect.

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- |psi(x)|^2 = [ (1 / (pi * a^2))^(1/4) ]^2 * exp( - x^2 / a^2 ) = ( 1 / (a * sqrt(pi)) ) * exp( - x^2 / a^2 ).
- Standard Gaussian integral: Integral_{-inf}^{+inf} exp(- u^2 ) du = sqrt(pi).
- Let u = x / a, dx = a * du:
Integral_{-inf}^{+inf} |psi(x)|^2 dx = ( 1 / (a * sqrt(pi)) ) * Integral_{-inf}^{+inf} exp(-u^2) * a du
= ( 1 / sqrt(pi) ) * sqrt(pi) = 1.000.
- The wave function is rigorously normalized.

2. Probability in the Interval -a <= x <= +a:
- P(-a <= x <= a) = ( 1 / sqrt(pi) ) * Integral_{-1}^{+1} exp(-u^2) du = ( 2 / sqrt(pi) ) * Integral_0^1 exp(-u^2) du.
- By definition of the Gauss Error Function: erf(z) = (2 / sqrt(pi)) * Integral_0^z exp(-u^2) du.
- P = erf(1.0).
- From mathematical tables: erf(1.0) ~= 0.8427 = 84.27%.

3. Physical Interpretation:
- The probability of detecting the nanoparticle within one standard deviation (+- a) of the origin is 84.27%.

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**>> Solution to Question 29:** Explain how the principles of quantum mechanics govern the hydration and setting of Portland cement at the sub-nanometer scale, with

**Solution to Question 30:** >>Please Synthesize the historical evolution of quantum mechanics and its profound transition from abstract 20th-century theoretical physics into